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Unit – 4 : Session – 8 : SLO – 1 | AVL Tree — Rotations (LL, RR, LR, RL)

■ I. LL and RR Rotations

- In an AVL tree, LL rotation is performed when a node becomes unbalanced due to insertion in the LEFT subtree of its LEFT child.
- LL Rotation is a type of SINGLE (right) rotation used to balance an AVL tree.
- After an LL rotation, the height difference between the left and right subtrees of the previously unbalanced node becomes at most 1.
- In an LL rotation, the left child of the unbalanced node becomes the new ROOT of the subtree.
- The time complexity of an LL rotation operation in an AVL tree is $O(1)$.
- RR rotation is used in an AVL tree when a node becomes unbalanced due to insertion in the RIGHT subtree of its RIGHT child.
- RR rotation is a type of SINGLE (left) rotation in AVL trees.
- In RR rotation, the right child of the unbalanced node becomes the new ROOT of the subtree.
- After an RR rotation, the height difference between the left and right subtrees of the previously unbalanced node becomes at most 1.
- The time complexity of an RR rotation operation in an AVL tree is $O(1)$.

■ II. LR and RL Rotations

- In an AVL tree, LR rotation is performed when the left child of a node is RIGHT heavy.
- LR rotation is a combination of LEFT (RR) rotation followed by RIGHT (LL) rotation.
- To fix an imbalance at a node caused by insertion in the right subtree of its left child, we perform an LR rotation.
- After an LR rotation, the node that was inserted becomes the ROOT of the subtree.
- In LR rotation, the imbalance occurs at node A, and its left child B has a right child C; the first step is a LEFT (RR) rotation on node B.
- RL rotation is required when a node becomes unbalanced due to insertion in the LEFT subtree of its RIGHT child.
- RL rotation is a combination of two rotations: first a RIGHT (LL) rotation on the right child, followed by a LEFT (RR) rotation on the unbalanced node.
- RL rotation is a type of DOUBLE rotation in AVL trees.
- In RL rotation, the node that becomes the new root of the subtree is the LEFT child of the right child of the unbalanced node.
- The main goal of performing an RL rotation is to restore the BALANCE (height-balance) property of the AVL tree.